INVESTIGATION BOOKLET • YEARS 7 AND 8

The Trebuchet Investigation

How does the counterweight change the throw?

Forces and Motion | Victorian Curriculum F–10 Version 2.0
VC2S8U13 | VC2S8U14 | VC2S8U15 | VC2S8I01–I08

Name: _____ **Class:** _____

Teacher: _____ **Date:** _____

My group: _____

How this investigation runs

You work in a group of four, but everyone fills in their own booklet. The investigation runs over four lessons:

- **Lesson 15** **Plan.** Part A, a reading about the trebuchet. Part B, the science of the swing, the sensor, your prediction, the variables and safety.
- **Lesson 16** **Launch.** The first half of Part C: five counterweights, three launches each.
- **Lesson 17** **Analyse.** The rest of Part C: graphs, analysis and conclusion. Then Part D.
- **Lesson 18** **Test.** An open-booklet test. Bring this booklet: some of the questions are about your own results.

Part D is above the Year 7 level. Try it once Parts A to C are finished.

Part	Title	Lesson	Page
A	Reading and comprehension — *Warwolf*	15	p. 2
B	Planning the investigation	15	p. 4
C	Results and analysis	16–17	p. 11
D	Above level	17	p. 17

Part A • Reading and comprehension

Reading — Warwolf

In the summer of 1304, the army of King Edward I of England surrounded Stirling Castle in Scotland. The castle stands on a steep crag of rock, and its defenders had already held out against Edward's other siege engines. So Edward ordered his engineers to build the biggest siege engine anyone had seen: a counterweight trebuchet called *Warwolf*. Five master carpenters and about fifty workers spent around three months building it.

Before Warwolf was finished, the defenders offered to surrender. Edward refused. He wanted to see his new machine in action, and he would not accept their surrender until Warwolf had thrown its first stones at the walls.

Why were people so impressed by the counterweight trebuchet? Earlier siege engines were powered by teams of people hauling on ropes. A tired team, or one that pulled out of step, pulled with a different force each time, so every shot landed somewhere different. A counterweight trebuchet replaced the people with gravity. A huge box of stones or earth — several tonnes, on the biggest machines — was winched up high and then let go. Gravity pulls on the same mass in the same way every time, so a well-built trebuchet could throw stone after stone onto almost the same spot, hammering one part of a wall until it collapsed.

Every stage of a trebuchet is physics you have already studied. Winching the counterweight up stores gravitational potential energy. When it is let go, an unbalanced force acts on it and it falls, turning that store into kinetic energy. The arm is a lever with its pivot close to the counterweight: the counterweight falls a short distance on the short arm, while the end of the long arm sweeps through a huge arc, much faster. A sling on the end of the arm whips the stone faster still. Once the stone is let go, the only forces on it are its weight and air resistance.

Medieval engineers improved their trebuchets by trial and error: a little more weight here, a longer sling there. In this investigation you will be the engineer, but you will work like a scientist. You will change one thing — the mass of the counterweight — measure what happens, and use a sensor to see inside the swing.

C1. Where and when did the siege in the reading take place, and who led the attacking army?

C2. The defenders offered to surrender before Warwolf was finished. Why did Edward refuse?

C3. Earlier siege engines were powered by teams of people. Explain why a counterweight trebuchet could hit the same spot again and again, when the earlier engines could not.

C4. Name the energy store of a raised counterweight, and the store it is turned into as the counterweight falls.

C5. The trebuchet's arm is a lever. Which end of the arm moves a short distance, and which end moves fast?

C6. In this investigation, what is the one thing you will change?

C7. Medieval engineers improved their trebuchets by trial and error. Suggest two ways this investigation will be more scientific than trial and error.

Part B • Planning the investigation

The Trebuchet Investigation

Curriculum: VC2S8U13 • VC2S8U14 • VC2S8U15 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05 • VC2S8I06 • VC2S8I07 • VC2S8I08

Question: *How does the mass of the counterweight affect how far a trebuchet throws, and how fast its arm swings?*

B1 How a Trebuchet Throws

A trebuchet is a **Class 1 lever**. Its arm turns on a **pivot** that sits between the counterweight and the projectile. The **counterweight** hangs from the end of the short arm. The **sling** hangs from the end of the long arm, with the projectile in its pouch. In this investigation a **sensor** is strapped to the arm as well.

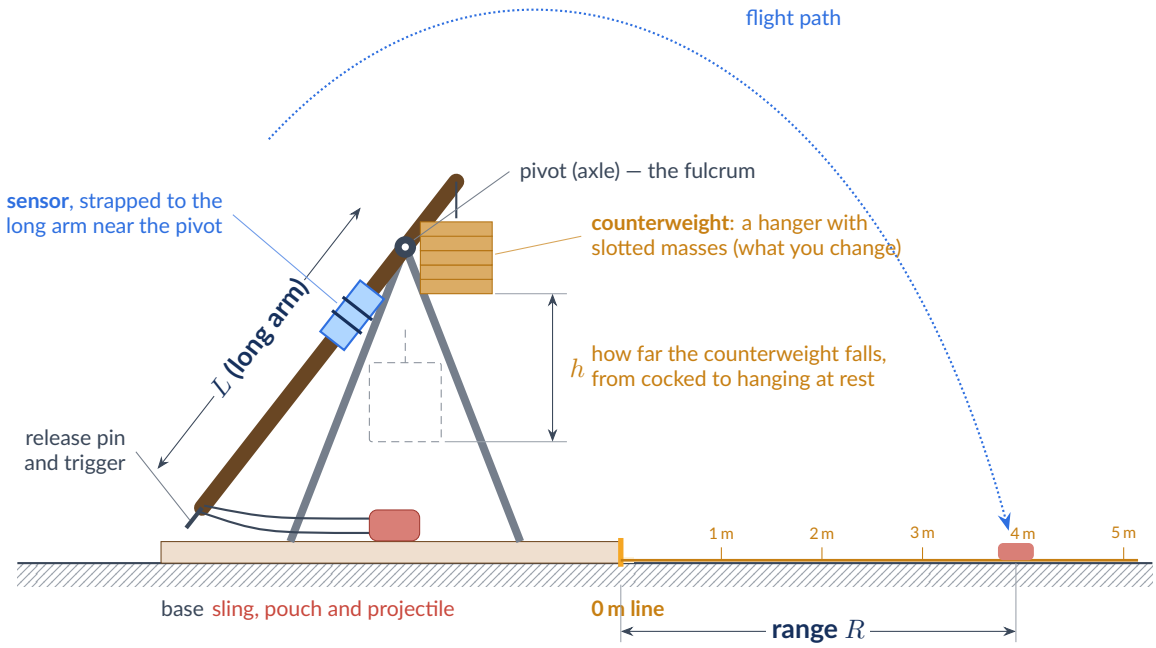
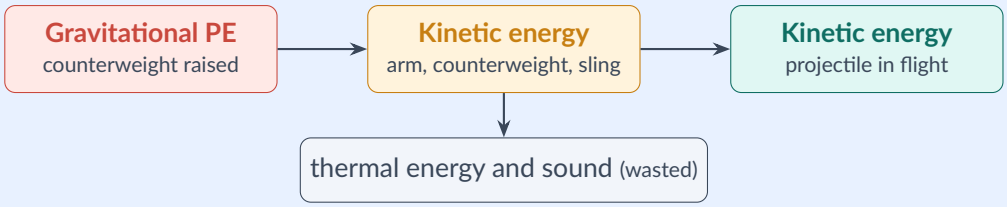


Figure 1 — A desktop trebuchet, cocked and ready. L is the length of the long arm, h is how far the counterweight falls, and the range R is measured from the 0 m line at the front of the base.

One launch, step by step

1. The trigger is released. Gravity is now an **unbalanced force** on the counterweight, so it falls.
2. As the short arm swings down, the long arm swings up and over the top. The whole arm turns about the pivot.
3. The sling is dragged along the base, then lifted and whipped around behind the arm.
4. When the sling has swung far enough, its loop slides off the release pin and the projectile flies free.
5. In flight, the only forces on the projectile are its **weight** and **air resistance**.

Where the energy goes



Raising the counterweight stores gravitational potential energy, GPE $= mgh$. If the counterweight always falls the same height h, then **doubling its mass doubles the energy it can give**. Some of the energy is wasted as thermal energy and sound, at the pivot and in the air.

Why the long arm moves fast

The pivot is close to the counterweight. While the counterweight falls a short distance, the end of the long arm sweeps through a much longer arc *in the same time*, so it moves much faster. A trebuchet gives away force to gain speed: its mechanical advantage is less than 1.

B2 Measuring the Swing with a Sensor

Your sensor, a PASCO Wireless 3-Axis Acceleration/Altimeter, contains a **gyroscope**: a tiny chip, like the one that lets a phone know it is being turned. It measures how fast the sensor is turning — its **spin rate**. Scientists call this *angular velocity*, and measure it in **radians per second (rad/s)**. One full turn is about 6.3 radians, so a spin rate of 6.3 rad/s is about one full turn every second.

Every point on the arm turns together

The arm is rigid, so every point on it turns through the same angle in the same time. That means the sensor reads the **same spin rate** wherever it is strapped on the arm — which is why it can sit safely near the pivot. But points further from the pivot travel around bigger circles in that time, so they **move faster**.

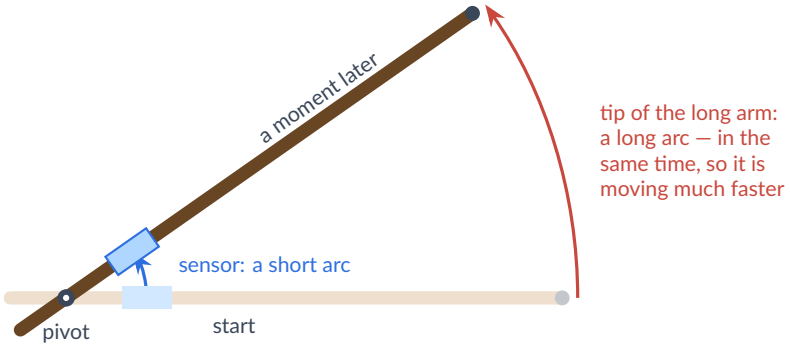


Figure 2 — The whole arm turns through the same angle, but the tip travels a much longer arc than the sensor.

Speed of the tip of the arm

$$v = \omega \times L$$

v = speed of the tip of the long arm (m/s) ω (say "omega") = spin rate from the sensor (rad/s)
L = length of the long arm, from the pivot to where the sling hangs, **in metres**

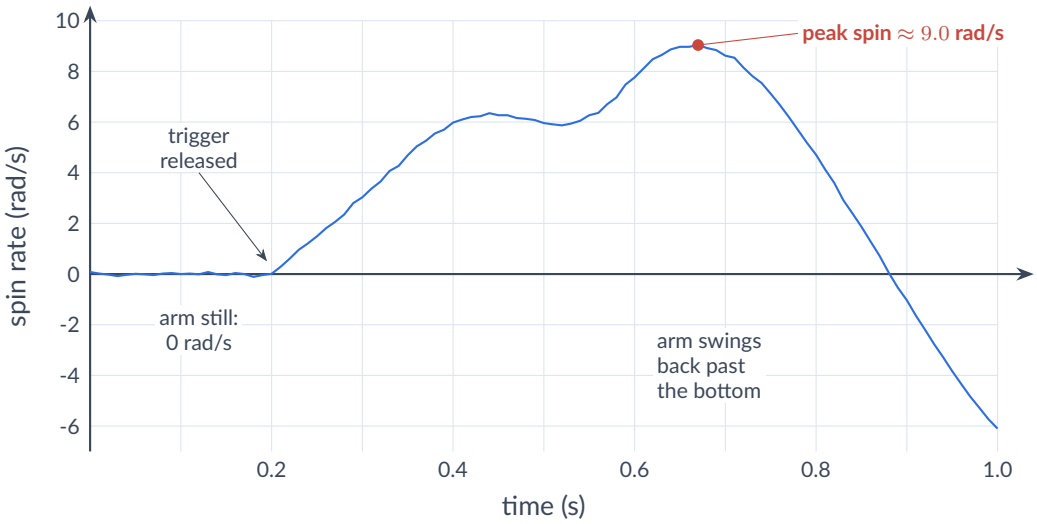


Figure 3 — What the sensor recorded during one launch with an 800 g counterweight.

Worked Example 1 — Peak spin and tip speed

The graph above is from a launch with an 800 g counterweight. The long arm is 0.40 m and the short arm is 0.10 m. Find the peak spin, the speed of the tip, and the speed of the counterweight end of the arm.

Peak spin: the highest point of the hump is about **9.0** rad/s.
Tip speed: $v = \omega \times L = 9.0 \times 0.40 =$ **3.6** m/s
Counterweight end: $v = 9.0 \times 0.10 =$ **0.9** m/s. The tip moves $3.6 \div 0.9 = 4$ times as fast — the same as long arm ÷ short arm $= 0.40 \div 0.10 = 4$.

Reading the sensor

- The sensor measures spin about three axes, x, y and z. Only one of them shows a big hump when the arm swings — the one lined up with the pivot's axle. Use that one.
- If the hump points downwards, the reading is negative. That only means the arm turned the other way round compared with the sensor. Use the size of the peak and ignore the minus sign.
- After the peak, the arm slows and swings back. That shows up as a dip below zero — ignore it.

Task 1

A launch gives a peak spin of 7.5 rad/s. The trebuchet's long arm is 0.50 m and its short arm is 0.10 m.

(a) Calculate the speed of the tip of the long arm.

(b) Calculate the speed of the counterweight end of the arm.

(c) How many times faster does the tip move than the counterweight end?

Task 2

Two groups strap the same sensor to the same trebuchet arm. Group 1 straps it 5 cm from the pivot. Group 2 straps it 20 cm from the pivot. Will their peak spin readings be the same or different? Explain your answer.

B3 Planning a Fair Test

You will change the **mass of the counterweight**, using five different masses with three launches at each. On every launch you will measure two things: the **range**, from the 0 m line to where the projectile first lands, and the **peak spin** of the arm, from the sensor. Everything else must stay the same.

Task 3

Complete the table for the variables in this investigation.

Variable	Name of the variable	How we change it or measure it	Unit
Independent (what we change)			
Dependent 1 (what we measure)			
Dependent 2 (what the sensor measures)			

Equipment (for each group)

- Desktop trebuchet with a counterweight hanger
- Slotted masses for five counterweights
- One beanbag projectile — the same one every launch
- PASCO acceleration sensor and its strap
- Tablet or laptop with SPARKvue
- Measuring tape (at least 8 m) and a metre ruler
- Masking tape, and a counter to mark the landing spot
- Safety glasses for everyone in the group

Group roles — swap roles after each counterweight

Operator: loads, cocks and releases the trebuchet. The only person beside it.
Safety officer: checks the range is clear, then calls “Launching!” and, afterwards, “All clear”.
Spotter: stands well to the side of the range and puts a counter where the projectile first lands.
Recorder: runs SPARKvue, reads the peak spin, and records both results.

Task 4

To make it a fair test, these things must stay the same on every launch. The first row is done for you. Complete the others, and add one more of your own in the last row.

Kept the same	If it changed, how could it change the results?	How we keep it the same
The projectile	*A heavier or lighter projectile would fly a different distance.*	*Use the same beanbag for every launch.*
The sling and release pin		
How far the arm is pulled back		
Where the sensor is strapped		
Where the range is measured from		

Task 5

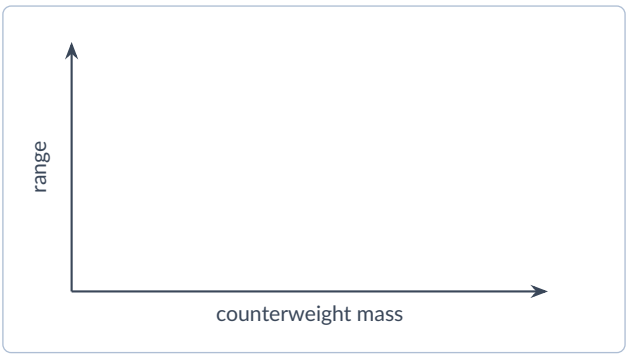
Make a prediction before you launch.

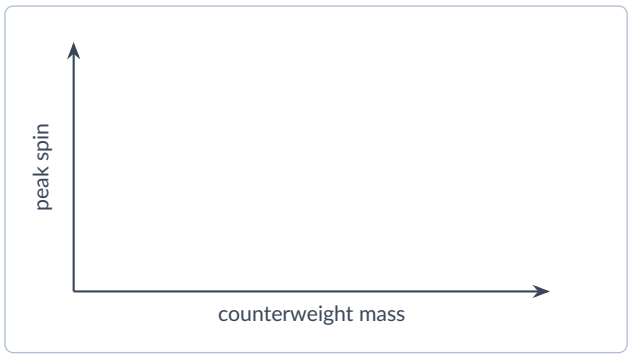
(a) Complete the prediction, and give a reason using energy or levers.

Prediction

I predict that as the mass of the counterweight increases, the range will _____ and the peak spin of the arm will _____, because …

(b) Sketch the shape you expect for each graph. You do not need any numbers.





Method

Setting up

1. Place the trebuchet on the floor, facing down the range. Stick a strip of masking tape on the floor along the front edge of the base. This is the **0 m line**. Line the base up with it before every launch.
2. Lay the measuring tape on the floor from the 0 m line, straight down the range.
3. Strap the sensor tightly to the long arm, as close to the pivot as you can while it still swings clear of the frame. Swing the arm slowly by hand to check. Do not move the sensor again.
4. Turn the sensor on and connect it in SPARKvue. Show a graph of angular velocity against time, at the sample rate your teacher gives you.
5. Measure the long arm L, the counterweight drop h and the mass of the projectile. Record them in Part C.

Launching

6. Load the hanger to make the smallest counterweight.
7. Pull the long arm down and set the trigger. Put the projectile in the pouch and hook the sling's loop over the release pin. Keep your hands out of the arm's path.
8. The safety officer checks that the range is clear and everyone is behind the trebuchet, then calls "Launching!"
9. The recorder presses **Start**. The operator releases the trigger. Press **Stop** once the arm has stopped swinging.
10. The spotter puts a counter where the projectile **first** landed. Measure from the 0 m line to the counter, to the nearest centimetre.
11. Read the peak spin from the graph. Record it with the range.
12. Launch three times with this counterweight. Then repeat steps 6 to 11 for each of the five counterweights.

Task 6

You will launch three times with each counterweight. Give two reasons why three launches are better than one.

Task 7

Complete the risk assessment. The first row is done for you.

Hazard	What could happen	How we will control the risk
The arm swings very fast.	*It could hit someone's head or hand.*	*Only the operator stands beside the trebuchet. Everyone else stays behind it.*
The projectile in flight		
The counterweight		

Safety rules

Safety glasses on while launching. • Never stand in front of the trebuchet or beside the range. • Soft beanbag projectiles only. • Collect projectiles only after "All clear". • Keep your hands out of the arm's path while it is cocked. • Change the counterweight only while the arm is hanging at rest, never while it is cocked.

Notes

Part C • Results and analysis

C1 Before You Launch

Task 8

Measure and record these before your first launch. The first row shows you how.

Measurement	Value (with unit)	How we measured it
e.g. Height of the pivot above the floor	*0.35 m*	*Metre ruler, from the floor to the centre of the axle*
Long arm length L (pivot to where the sling hangs)		
Counterweight drop h (cocked to hanging at rest)		
Mass of the projectile		
Sensor sample rate		

Task 9

Decide who does each job. Swap jobs for every new counterweight, so that everyone does every job at least once. The first row is an example.

Counter-weight (g)	Operator	Safety officer	Spotter	Recorder
e.g. 500	*Ava*	*Ben*	*Chloe*	*Dev*

C2 Results

Task 10

Record the range of every launch, then calculate the mean range for each counterweight. The first row is an example.

Counterweight mass (g)	Range, launch 1 (m)	Range, launch 2 (m)	Range, launch 3 (m)	Mean range (m)
e.g. 500	*1.64*	*1.71*	*1.75*	*1.70*

Task 11

Record the peak spin of every launch and its mean. Then find the tip speed with $v = \omega \times L$, using your long arm in metres. The example is for a 0.40 m arm.

Counterweight mass (g)	Peak spin, launch 1 (rad/s)	Peak spin, launch 2 (rad/s)	Peak spin, launch 3 (rad/s)	Mean peak spin (rad/s)	Tip speed $v = \omega \times L$ (m/s)
e.g. 500	*7.9*	*8.1*	*8.0*	*8.0*	$8.0 \times 0.40 = 3.2$

Means and odd results

Mean: add the three launches and divide by 3. For 1.64, 1.71 and 1.75 m: $(1.64 + 1.71 + 1.75) \div 3 = 5.10 \div 3 =$ **1.70** m.

Odd result: if one launch is very different from the other two — the sling let go too early, or the projectile hit something — circle it, write down what happened, and launch again. Leave the odd result out of the mean.

Task 12

Did any launch give a result very different from the others with the same counterweight? Describe it and explain what you think went wrong. Was the peak spin odd for that launch too? What does that tell you?

C3 Graphs

Task 13

Draw a line graph of **mean range** (vertical axis) against **counterweight mass** (horizontal axis). Start both axes at zero, choose scales that use most of the grid, plot your five points, draw a smooth line of best fit, and give the graph a title.

Task 14

Draw a second line graph, of **tip speed** (vertical axis) against **counterweight mass** (horizontal axis), in the same way. Start both axes at zero again, so that you can compare the two graphs fairly.

C4 Analysis

Task 15

Describe the pattern in your range graph, using numbers from your results.

Sentence starter

As the counterweight mass increased from _____ g to _____ g, the mean range increased / decreased from _____ m to _____ m. The line of best fit was straight / curved, which means …

Task 16

Compare your heaviest and your lightest counterweights.

(a) How many times further did the projectile go? Divide the mean range for the heaviest by the mean range for the lightest.

(b) How many times faster did the tip of the arm move? Divide the two tip speeds in the same way.

(c) Which changed by more: the range, or the speed of the arm?

Task 17

Explain *why* a heavier counterweight throws the projectile further. Use at least four of the words in the word bank.

gravitational potential energy | kinetic energy | unbalanced force | lever | pivot | speed | mass

Task 18

Look at the three launches for each counterweight in your range table.

(a) Which counterweight had the biggest spread between its three launches? How big was the spread (largest range − smallest range)?

(b) Were your results repeatable? Explain using your numbers.

Task 19

Identify sources of error in your investigation, and suggest how to reduce each one. The first row is an example.

Source of error	How it affected the results	How to reduce it
The spotter judged the landing spot by eye.	*Ranges could be a few centimetres out.*	*Film the landing, or have two spotters agree on the spot.*

Task 20

Write down the prediction you made before launching. Then say whether your results support it, using numbers as evidence.

Sentence starter

I predicted that … My results showed that … This supports / does not support my prediction because …

C5 Conclusion

Task 21

Write a conclusion that answers the investigation question: *How does the mass of the counterweight affect how far a trebuchet throws, and how fast its arm swings?* Include the pattern you found with numbers, a scientific reason for it, how reliable your results were, and one improvement for next time.

Sentence starters

This investigation found that … For example, … This happens because … Our results were (were not) reliable because … Next time we would …

Part D • Above level

These questions go beyond the Year 7 standard. Each one gives you the data it needs, all from one group's investigation with a 25 g projectile and a long arm of 0.40 m. Use $g = 9.8$ N/kg.

Extension 22

Launch speed from range. When a projectile is launched at about $45°$, its launch speed is roughly

$$v_{\text{launch}} \approx \sqrt{g \times R} \qquad \text{where } R \text{ is the range in metres.}$$

(a) Complete the table. The first row is done for you.

Counter-weight (g)	Mean range R (m)	Launch speed $\sqrt{g \times R}$ (m/s)	Tip speed of the arm (m/s)	Launch speed ÷ tip speed
800	*4.18*	$\sqrt{9.8 \times 4.18} = 6.4$	*3.6*	$6.4 \div 3.6 = 1.8$
600	2.49		3.3	
1000	5.75		3.8	

(b) The projectile leaves faster than the tip of the arm. Explain how the sling makes this possible. (Hint: think of the sling as an extra piece of arm.)

(c) With the heavier counterweight the projectile was launched much faster, but the tip of the arm was only a little faster. Suggest why.

Extension 23

How efficient is the trebuchet? The 800 g counterweight falls $h = 0.18$ m. The 25 g projectile is launched at 6.4 m/s.

(a) Calculate the gravitational potential energy the counterweight gives up: GPE $= mgh$, with the mass in kilograms.

(b) Calculate the kinetic energy of the projectile: KE $= \frac{1}{2}mv^2$.

(c) Calculate the efficiency: useful energy out ÷ energy in $\times 100\%$.

(d) Name two places the rest of the energy went.

Extension 24

The limit on the arm. The group's mean peak spin rose from 8.2 rad/s with 600 g to only 9.5 rad/s with 1000 g.

(a) Calculate the percentage increase in the peak spin.

(b) A student says: "With a counterweight a hundred times heavier, the arm would spin a hundred times faster." Explain why this cannot be true. (Hint: could the counterweight ever fall faster than an object simply dropped from the same height?)

Extension 25

Weightless for half a second. The sensor also measures acceleration. Sitting still on the bench, it reads about 9.8 m/s^2. Your teacher holds it 1.00 m above a cushion and lets go. While it falls, the reading drops to almost 0 for 0.45 s, then jumps when it hits the cushion.

(a) Draw a labelled free body diagram of the sensor sitting on the bench, and one of the sensor falling.

Sitting on the bench

Falling

(b) The sensor can only feel forces from things that touch it. Use your diagrams to explain why it reads almost 0 while it falls, even though gravity is still pulling on it.

(c) For an object dropped from rest, $g = 2h \div t^2$. Use this to calculate g from the drop.